

Practice Problem Set 2

ECON 300 – Labour Economics

Winter 2026

Problem 1:

Consider an income maintenance program with the following structure:

- Guaranteed income (benefit): $G = 600$ per week
- Benefit reduction rate (tax-back rate): $t = 0.5$
- Market wage: $w = 20$ per hour

Disposable income is given by:

$$C = G + (1 - t)wh \quad \text{for } wh \leq \frac{G}{t}$$

1. Calculate the **break-even level of earnings** where benefits fall to zero.
2. Compute the worker's **effective net wage** while receiving benefits.
3. Compare the net wage to the market wage and explain the implication for labour supply incentives.
4. Explain why high benefit reduction rates may discourage work even when market wages are unchanged.

Problem 2:

A worker has 80 hours per week available for work and leisure.

1. Draw the budget constraint under:
 - No income support
 - A guaranteed income program with a benefit reduction rate less than 100%
2. Clearly label:
 - The guaranteed income level
 - The break-even point

3. Explain how the slope of the budget constraint changes when moving from the no-program case to the program case.
4. Using indifference curves, explain how the program may affect:
 - Labour force participation
 - Hours worked by individuals who remain employed

Problem 3:

Answer each question briefly (3–5 sentences each).

1. How does a **negative income tax** differ from a traditional welfare program?
2. Why may an **earned income tax credit (EITC)** increase labour force participation but reduce hours worked for some individuals?
3. Why do economists distinguish between the **extensive margin** and the **intensive margin** of labour supply?
4. Why does targeting income support to low-income workers create trade-offs between equity and efficiency?

Problem 5:

A competitive firm has the production function:

$$Q = 10L^{0.5}$$

where L is labour.

The firm faces:

- Output price: $P = 4$
 - Wage rate: $w = 8$
1. Derive the firm's **marginal product of labour (MPL)**.
 2. Derive the **value of marginal product of labour (VMPL)**.
 3. Determine the firm's **optimal employment level**.
 4. Explain why the labour demand curve is downward sloping in the wage.

Problem 6:

1. Draw the firm's labour demand curve derived from the VMPL.
2. Show equilibrium employment when:
 - The wage is $w = 8$
 - The wage increases to $w = 12$
3. Explain how a change in the wage differs from a change in:
 - Output price
 - Technology

Problem 7:

Using the production function from Problem 5:

1. Recalculate the firm's optimal employment level if the output price increases from $P = 4$ to $P = 6$.
2. Explain the result intuitively using the concept of the value of marginal product.
3. Distinguish clearly between:
 - A movement along the labour demand curve
 - A shift of the labour demand curve

End of Practice Problem Set